

JEE(Main + Advanced) : NURTURE COURSE (PHASE : S(TEPS))
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,C	C,D	B,D	C,D	A,C,D	A,C	B,C,D	A,C,D	A,D
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	5	2	5	8	8				
SECTION-IV	Q.	1	2								
	A.	A->P,B->Q,C->R,D->P	A->PR,B->P,C->PT,D->PS								

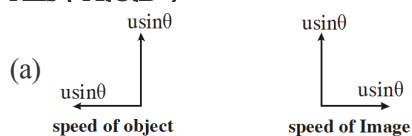
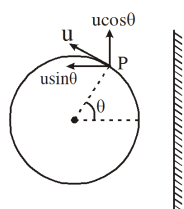
PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,C,D	A,C	A,B,D	A,C	A,B,D	A,B	A,B,C,D	B,D	B,D
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	6	5	5	3	4				
SECTION-IV	Q.	1	2								
	A.	A->PRST,B->PQRST,C->PQRST,D->PRST	A->PQR,B->RS,C->RT,D->R								

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C,D	A,C	A,B,C,D	A,B	A,B,C	B,C	A,C,D	B,C,D	D
SECTION-III	Q.	1	2	3	4	5	6				
	A.	2	0	2	4	6	5				
SECTION-IV	Q.	1	2								
	A.	A->Q,B->P,C->S,D->R	A->Q,B->PR,C->PST,D->Q								

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (A,C,D)**

 relative speed = $2 u \sin \theta$ (perpendicular to mirror)

 (b) relative speed = $0 \Rightarrow \theta = 0, 180^\circ$ since relative velocity is perpendicular to the mirror

 (c) $\therefore$ path will be to and fro along a straight line perpendicular to the mirror.

 2. **Ans (A,C)**

Due to the introduction of glass slot object displaces by an amount of $t \left[1 - \frac{1}{\mu} \right] = 3 \left[1 - \frac{2}{3} \right] \Rightarrow 1\text{cm}$ in the direction of the ray of incident ray.

 4. **Ans (B,D)**

$$\vec{V}_{s/G} = \vec{V}_{s/A} + \vec{V}_{A/G}$$

Horizontal component of velocity of shell will be in the direction of velocity of aeroplane

5. Ans (C,D)

$$F_1 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} = 50\text{Hz}$$

$$F_2 = 100\text{ Hz}$$

$$F_3 = 150\text{ Hz}$$

$$n_1 = \frac{1}{4\ell} \sqrt{\frac{T}{\mu}} = 25\text{Hz}$$

$$n_2 = 75\text{ Hz}$$

$$n_3 = 125\text{ Hz}$$

8. Ans (B,C,D)

$$U_i + K_i = U_f + K_f$$

$$2kq\lambda\ell\ln\left(\frac{a}{R/2}\right) + 0 = 2kq\lambda\ell\ln\left(\frac{a}{3R/2}\right) + \frac{1}{2}ma^2$$

$$v_{\max} = \sqrt{\frac{q\lambda}{\pi\epsilon_0} \frac{\ell n3}{m}}$$

9. Ans (A,C,D)

$$\frac{kQ_a}{(\ell + x_1)^2} = \frac{kQ_b}{x_1^2}$$

$$\frac{Q_a}{Q_b} = \left(\frac{\ell + x_1}{x_1}\right)^2$$

$$E = \frac{kQ_a}{(\ell + x)^2} - \frac{kQ_b}{x^2}$$

$$\text{at max } \frac{dE}{dx} = 0 = -\frac{2kQ_a}{(\ell + x)^3} + \frac{2kQ_b}{x^3}$$

$$\Rightarrow \frac{Q_a}{Q_b} = \left(\frac{\ell + x_2}{x_2}\right)^3 = \left(\frac{\ell + x_1}{x_1}\right)^2$$

$$\frac{\ell}{x_2} + 1 = \left(\frac{\ell + x_1}{x_1}\right)^{2/3}$$

$$\Rightarrow x_2 = \frac{\ell}{\left(\frac{\ell + x_1}{x_1}\right)^{2/3} - 1}$$

10. Ans (A,D)

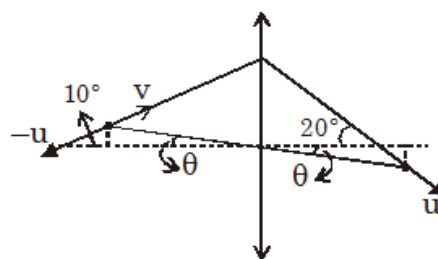


Figure (1)

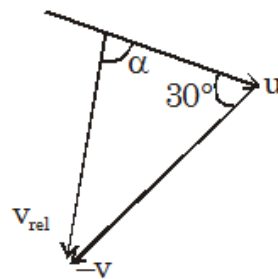


Figure (2)

$$\frac{u_{\text{rel}}}{\sin 30^\circ} = \frac{u}{\sin \alpha}$$

$$u_{\text{rel}} = \frac{u \sin 30^\circ}{\sin \alpha}$$

$$u_{\text{rel}} = \text{min if } \alpha = 90^\circ$$

$$u_{\text{rel min}} = \frac{u}{2}$$

$$v_{1x} = +m^2 v_0 x$$

$$u \cos 20^\circ = m^2 \times v \cos 10^\circ$$

$$\text{also } v \cos 30^\circ = u$$

$$\frac{\cos 30^\circ \cos 20^\circ}{\cos 10^\circ} = m^2$$

$$m^2 = \frac{[\cos 50^\circ + \cos 10^\circ]}{2 \cos 10^\circ}$$

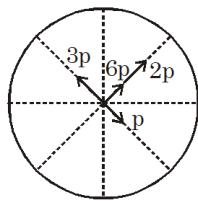
$$= \frac{\cos 50^\circ}{2 \cos 10^\circ} + 2$$

$$= \frac{\cos 60^\circ \cos 10^\circ + \sin 60^\circ \sin 10^\circ}{2 \cos 10^\circ} + 2$$

$$= \frac{1}{4} + \frac{\sqrt{3}}{4} \times \frac{10\pi}{180} + 2$$

SECTION-III

1. Ans (4)



$$p_{+q} = \frac{2\sqrt{2}qR}{\pi} = p$$

$$p_{+2q} = \frac{2\sqrt{2}(2q)R}{\pi} = 2p$$

$$p_{+3q} = \frac{2\sqrt{2}(3q)R}{\pi} = 3p$$

$$p_{-6q} = \frac{2\sqrt{2}(6q)R}{\pi} = 6p$$

$$p_{eq} = \sqrt{68}p = \sqrt{68} \frac{2\sqrt{2}(qR)}{\pi}$$

2. Ans (5)

$$1.500 = \frac{330}{330 - a \times 22}$$

$$330 - 22a = 220$$

$$a = 5 \text{ m/s}^2$$

3. Ans (2)

$$\frac{1}{V} = CT^2$$

$$T^2V = C_1$$

$$TV^{1/2} = C_2$$

$$TV^{r-1} = \text{constant}$$

$$r = 3/2$$

$$V_{rms} = \sqrt{\frac{3RT}{M}}, V_{sound} = \sqrt{\frac{\gamma RT}{M}}$$

$$\frac{V_{rms}}{V_{sound}} = \sqrt{\frac{3}{\gamma}} = \sqrt{2}$$

4. Ans (5)

$$-7T + 9T - 2T \cdot V_B = 0$$

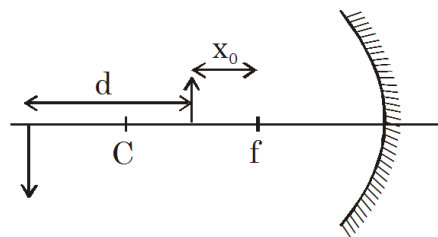
$$2 = 2V_B$$

$$V_B = 1 \text{ m/s}$$

$$-2T - 3T + 2T \cdot a_B = 0$$

$$a_B = \frac{5}{2} = 2.5 \text{ m/s}^2$$

5. Ans (8)



$$m = -\frac{f}{x_0} = -2$$

$$x_0 = \frac{f}{2}$$

$$x_0 x_I = f^2$$

$$x_0 (12 + x_0) = f^2$$

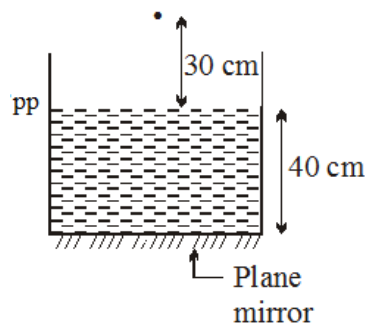
$$\frac{f}{2} \left(12 + \frac{f}{2} \right) = f^2$$

$$12 + \frac{f}{2} = 2f$$

$$12 = \frac{3f}{2}$$

$$f = 8 \text{ cm}$$

6. Ans (8)



$$v_0 = -4 \text{ m/s} \quad \downarrow \text{-ve}$$

$$v_{i/m} = -\mu v_0$$

$$v_{i/m} = +4/3 \times 4$$

$$v_{i/m} = +16/3$$

$$v_{i/G} = \frac{v_{i/m}}{\mu}$$

$$v_{i/G} = \frac{+16/3}{4/3} = 4$$

$$v_{i/0} = 4 - (-4) = 8 \text{ m/s}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,B)**

$$a\ell^2 + b m^2 + 2d\ell + 1 = 0$$

$$a = b + d^2$$

$$(b + d^2)\ell^2 + b m^2 + 2d\ell + 1 = 0$$

$$\Rightarrow b(\ell^2 + m^2) + \ell^2 d^2 + 2d\ell + 1 = 0$$

$$-b = \frac{(\ell d + 1)^2}{\ell^2 + m^2} \Rightarrow \left| \frac{\ell d + 1}{\sqrt{\ell^2 + m^2}} \right| = \sqrt{-b}$$

$$\text{Centre} \equiv (d, 0) \quad \text{radius} = \sqrt{-b}$$

$$\text{Circle : } (x - d)^2 + y^2 = -b$$

(C) tangent from origin is possible if $d^2 + b \geq 0$

2. **Ans (A,C,D)**

$$\text{Let } a = \frac{\alpha}{r}, \quad b = \alpha, \quad c = \alpha r$$

$$abc = 6^6$$

$$\therefore \alpha^3 = 6^6 \Rightarrow \alpha = 36$$

$$\text{Now } a \text{ is positive integer, } a = \frac{36}{r}$$

possible values of $r = 2, 3, 4, 6, 12, 18, 36$

Now $r = 4$ is only possible value of which

$$b - a = 36 \left(\frac{r-1}{r} \right) \text{ is perfect cube.}$$

Hence $a = 9, b = 36, c = 144$.

3. **Ans (A,C)**

Case-1 $\ell = m$

$$\cos A = \frac{7^2 + \ell^2 - \ell^2}{2 \cdot 7 \cdot \ell}$$

$$\ell = \frac{49}{4}$$

Hence sides are (Rejected)

Case-2 If $m = 7$

$$\cos A = \frac{7^2 + \ell^2 - m^2}{2 \cdot 7 \cdot \ell}$$

$$4\ell = \ell^2 \quad \therefore \ell = 4 \quad \text{sides are } 7, 7, 4$$

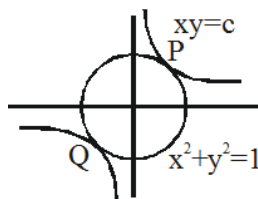
Case-3 One more case, which too is rejected

Using sides as 7, 7, 4

Area and circumradius are respectively $6\sqrt{5}$

$$\text{and } \frac{49\sqrt{5}}{30}$$

4. **Ans (A,B,C,D)**



To find point of contact solve both the curves.

$$x^2 + \frac{c^2}{x^2} = 1 \Rightarrow x^4 - x^2 + c^2 = 0$$

both touch each other $\Rightarrow D = 0$

$$1 - 4c^2 = 0 \Rightarrow c = \frac{1}{2} \quad (c > 0)$$

$$\Rightarrow \left(x^2 - \frac{1}{2} \right)^2 = 0$$

$$\Rightarrow x = \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$$

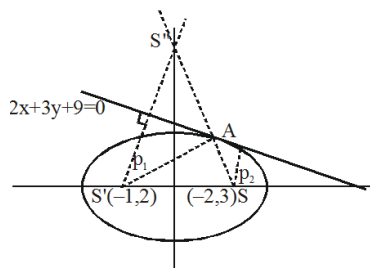
$$P \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), Q \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow PQ = 2$$

$$\text{Slope of tangent} \Rightarrow -\frac{(1/\sqrt{2})}{(1/\sqrt{2})} = -1$$

Length of latus rectum of $xy = c$ is 2

5. **Ans (A,B)**



$$p_1 = \left| \frac{-2 + 6 + 9}{\sqrt{13}} \right| = \sqrt{13}$$

$$p_2 = \left| \frac{14}{\sqrt{13}} \right| = \frac{14}{\sqrt{13}}$$

for an ellipse, $p_1 p_2 = b^2 \Rightarrow b^2 = 14$

$$L_{mA} = 2b = 2\sqrt{14} \quad (\text{A is correct})$$

$SS' = \sqrt{1+1} = \sqrt{2} = \text{Dist. between foci}$ (C is incorrect)

$p_1 p_2 = b^2 = 14$ for any tangent (D is incorrect)

$$\text{Now } SS' = \sqrt{2} = 2ae \Rightarrow 4a^2 e^2 = 2 \Rightarrow 2(a^2 - b^2) = 1$$

$$\Rightarrow a^2 = -14 + \frac{1}{2} = \frac{29}{2}$$

Image of $S'(-1,2)$ in the line $2x + 3y + 9 = 0$ is

$$S''(-5,-4)$$

6. **Ans (A,B,C)**

Number of positive divisors $= 4 \times 3 \times 5 = 60$
divisors which are divisible by 15 $= 4 \times 2 \times 4 = 32$
number of ordered pair (a,b) = number of divisors = 60

7. **Ans (B,C)**

$$N = \sum_{r=0}^{100} (-1)^r \cdot 4^{100-r} \cdot {}^{200}C_r \cdot {}^{200-r}C_{100-r}$$

$$= \sum_{r=0}^{100} (-1)^r \cdot 4^{100-r} \cdot \frac{|200|}{|r|} \cdot \frac{|200-r|}{|100-r|} \cdot \frac{|100|}{|100|}$$

$$= {}^{200}C_{100} \sum_{r=0}^{100} (-1)^r \cdot 4^{100-r} \cdot {}^{100}C_r$$

$$N = 3^{100} \cdot {}^{200}C_{100}$$

8. **Ans (A,C,D)**

Equation of tangent to $y^2 = 4x$ at (4, 4) is $x - 2y + 4 = 0$
Now, distance of the center of the variable circle from this tangent is equal to its distance from (1, 0). Hence, locus of the center will be a parabola having focus at (1, 0) & $x - 2y + 4 = 0$ as the directrix.

So, length of latus rectum

$$= 2 \times \frac{5}{\sqrt{5}} = 2\sqrt{5}$$

9. **Ans (B,C,D)**

$$\frac{x^2}{25} + \frac{y^2}{24} = 1, a = 5, b = 2\sqrt{6}, e = \frac{1}{5}$$

So, the point is on the directrix $x = \frac{a}{e} = 25$.

So AB is a focal chord, also as $\frac{1}{SA} + \frac{1}{SB} = \frac{2a}{b^2}$

$$\Rightarrow \text{if } S_1A = \frac{60}{13} \text{ then } S_1B = 5. \text{ Also } S_1A + S_2A = 2a = 10 = S_1B + S_2B$$

$$\Rightarrow S_2A = \frac{70}{13}, S_2B = 5. \text{ Distance from directrix} = \frac{1}{e} \times \text{distance from focus} = \frac{70}{13} \times 5 = \frac{350}{13}$$

10. **Ans (D)**

$$\sqrt{(x+3) - 2.2\sqrt{x-1}} + \sqrt{(x+15) - 2.4\sqrt{x-1}} = 2$$

$$|\sqrt{x-1} - 2| + |\sqrt{x-1} - 4| = 2$$

$$\text{Let } \sqrt{x-1} = t$$

$$|t-2| + |t-4| = 2$$

Case-I :

$$\text{If } 0 \leq t < 2$$

$$2 - t + 4 - t = 2$$

$$t = 2$$

Case-II :

$$2 \leq t < 4$$

$$t - 2 + 4 - t = 2$$

$$2 = 2$$

Case-III :

$$t \geq 4$$

$$t - 2 + t - 4 = 2$$

$$t = 4$$

$$\Rightarrow t = [2, 4]$$

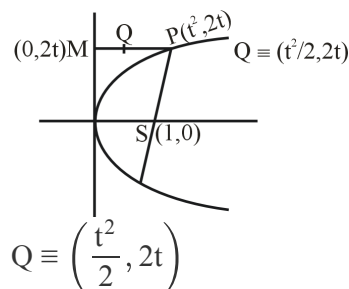
$$\sqrt{x-1} = [2, 4]$$

$$x - 1 \in [4, 16]$$

$$x \in [5, 17] \quad \text{infinite values}$$

SECTION-III

1. **Ans (2)**



Equation of PS:

$$(y-0) = \frac{2t}{t^2-1}(x-1)$$

$$\text{PS: } 2tx - (t^2-1)y - 2t = 0$$

Perpendicular distance from Q to PS

$$P = \left| \frac{t^3 - 2t(t^2-1) - 2t}{\sqrt{4t^2 + (t^2-1)^2}} \right| = \frac{t^3}{t^2+1}$$

On solving perpendicular distance $QL = f(t) = \frac{t^3}{t^2+1}$ so,

maximum value of $f(t)$ will be at $t=3$ which is $\frac{27}{10}$ units

2. Ans (A $\rightarrow$ Q, B $\rightarrow$ PR, C $\rightarrow$ PST, D $\rightarrow$ Q)

(A) Slope of line in new position is

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

Equation of line is

$$y - 3 = \frac{1}{\sqrt{3}}(x - 5)$$

$$x - \sqrt{3}y + 3\sqrt{3} - 5 = 0$$

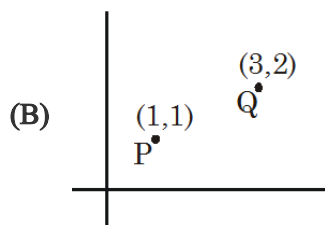
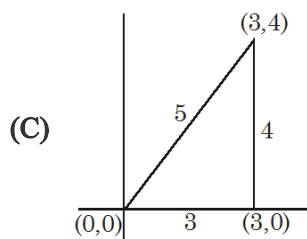


Image of P about x-axis is P'(1, -1)

Line joining P' and Q is $3x - 2y - 5 = 0$

Point on x-axis is $\left(\frac{5}{3}, 0\right)$

$$\therefore a + b + c = 8$$



$$I = \left[\frac{3(3) + 5(3) + 4(0)}{12}, \frac{3(4) + 5(0) + 4(0)}{12} \right]$$

$$I = [2, 1]$$

$$\therefore \alpha = 2, \beta = 3$$

(D) $x^4 - 2x^3 - x^2 - 2x + 1 = 0$

$$x^2 + \frac{1}{x^2} - 2 \left[x + \frac{1}{x} \right] - 1 = 0$$

$$x + \frac{1}{x} = y$$

$$y^2 - 2 - 2y - 1 = 0$$

$$y^2 - 2y - 3 = 0$$

$$(y - 1)^2 = 4$$

$$y - 1 = 2, -2$$

$$y = 3, -1$$

$$x + \frac{1}{x} = 3$$

$$\therefore x^2 - 3x + 1 = 0$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

JEE(Main + Advanced) : NURTURE COURSE (PHASE : S(TEPS))
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B	A,B	A,B	A,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A,C	A,B,C	A,C,D	A,C,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.60	1.73	0.05	22.36	1.12 to 1.13	11.00	24.00	4.00

PART-2 : CHEMISTRY

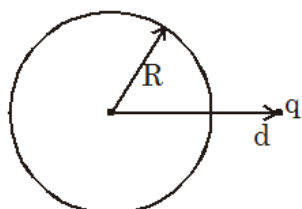
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	B,C,D	A,B,C,D	A,B,C	A,B,C	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A,D	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	5.00	6.00	4.00	1.40	0.78	0.50	14.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B,C	B,C,D	A,B,C	A,B	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B,C,D	A,B,D	C,D	A,C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.33	2.25	0.80	2.33	1.75	6.00	1.80	0.50

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

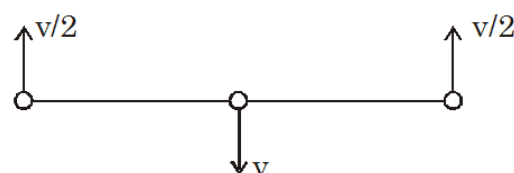
1. Ans (A,D)



$$E_q + E_{sp} = 0$$

$$\Rightarrow E_{sp} = \frac{kq}{d^2}$$

2. Ans (A,B)



$$\frac{kQ^2}{L} + 0 = \frac{kQ^2}{2L} + \frac{1}{2}mv^2 + \frac{1}{2}m\left(\frac{v}{2}\right)^2$$

$$\frac{kQ^2}{2L} = \frac{1}{2}m \times \frac{3v^2}{L}$$

$$v = \sqrt{\frac{Q^2}{6\pi\epsilon_0 mL}}$$

$$T - \frac{kQ^2}{L^2} - \frac{kQ^2}{4L^2} = \frac{m}{L} \left(\frac{3v}{2}\right)^2$$

$$T = \frac{5kQ^2}{4\pi^2} + \frac{9}{4} \times \frac{Q^2}{4\pi\epsilon_0 L^2} = \frac{Q^2}{4\pi\epsilon_0 L^2} \left[\frac{5}{4} + \frac{3}{7} \right]$$

3. Ans (A,B)

$$32.5 = \frac{kQ}{r^2} = \frac{9 \times 10^9 \times 130 \times 10^{-9}}{(2-x)^2 - (3+y)^2}$$

$$45 = \frac{9 \times 10^9 \times 130 \times 10^{-9}}{(3-x)^2 + (2-y)^2}$$

$$(2-x)^2 + (3+y)^2 = 36$$

$$4 - 4x + x^2 + 9 + 6y + y^2 = 36$$

$$9 + 6x + x^2 + 4 - 4y + y^2 = 26$$

$$x^2 + y^2 + 6y - 4x = 23$$

$$x^2 + y^2 - 4y + 6x = 13$$

$$10y - 10x = 10$$

$$y - x = 1$$

$$y = x + 1$$

$$9 + 6x + x^2 + 4 - 4x + (x^2 + 2x + 1) = 26$$

$$2x^2 + 4x - 16 = 0$$

$$x^2 + 2x - 8 = 0$$

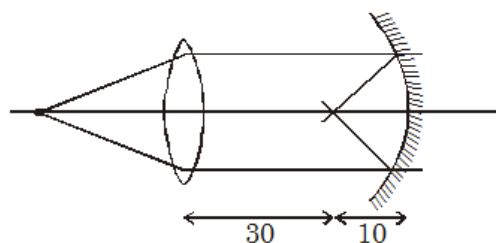
$$x = \frac{-2 \pm \sqrt{4 + 32}}{2} = -4, 2$$

$$y = -3, 3$$

5. Ans (A,D)

(A) Object for lens is between f & $2f$.

Image is beyond $2f$.

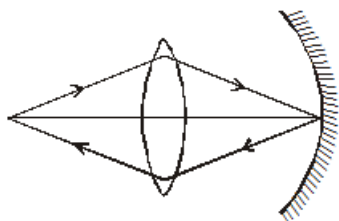


(B) For auto collimation object for mirror should be at its C is at 20 cm from mirror.

In B case, image is virtual not possible

In C case, rays are parallel not possible

In D case, if object is at 40 cm, image is formed at pole.



SECTION-I (ii)

8. Ans (A,B,C)

$$\frac{1}{2}mv^2 - \frac{Ze^2}{4\pi\epsilon_0 r_0} = 0$$

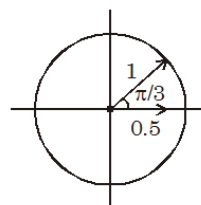
$$v = \sqrt{\frac{2Ze^2}{4\pi m\epsilon_0 r_0}}$$

$$3 \times 10^8 = \sqrt{\frac{2 \times Z \times 2.56 \times 10^{-38} \times 9 \times 10^9}{r_0 \times 9 \times 10^{-31}}}$$

$$9 \times 10^{16} < 200 \times 2.56 \frac{Z}{r_0}$$

$$\frac{Z}{r_0} < \frac{9 \times 10^{16}}{512}$$

9. Ans (A,C,D)



$$\frac{\pi}{3} = \omega t = 2\pi \times 400t$$

$$t = \frac{1}{2400} \text{ sec}$$

$$P = +ve \Rightarrow \vec{F} \cdot \vec{v} > 0$$

$$Fv \cos 0^\circ > 0$$

Since v is $+x$ at $t = 0$

$$v = \sqrt{\frac{N}{P}} = f\lambda$$

$$\lambda = \frac{1}{400} \sqrt{\frac{10^{11}}{7840}} = \frac{10^5}{400 \times 28} = \frac{125}{14}$$

10. Ans (A,C,D)

$$(A) p = B \frac{\partial s}{\partial x}$$

$$v_p = -v \frac{\partial s}{\partial x}$$

$$\frac{v_p}{p} = \frac{v}{B}$$

$$\text{if } v_p > 0 \Rightarrow p > 0$$

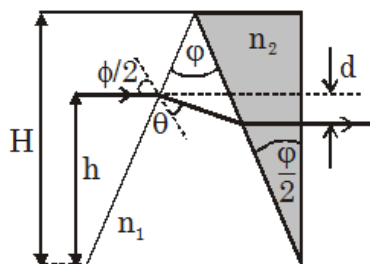
(B) & (C) at mean PE is maximum

$$v_{\max} = A\omega$$

$$= 10^{-3} \times 2\pi \times 400 = \frac{4\pi}{5} \text{ m/s}$$

SECTION-II

1. Ans (1.60)



$$1 \times \frac{\phi}{2} = 1.3 \times Q$$

$$\theta = \frac{2}{1.3}$$

$$\theta + \alpha = \phi = 4$$

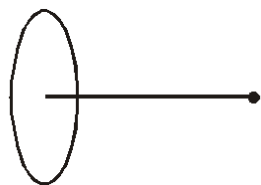
$$\alpha = 4 - \frac{2}{1.3} = \frac{3.2}{1.3}$$

$$n_1 \times \alpha = n_2 \times \frac{\phi}{2}$$

$$1.3 \times \frac{3.2}{1.3} = n_2 \times \frac{4}{2}$$

$$n_2 = 1.6$$

2. Ans (1.73)



$$E = \frac{kQx}{(R^2 + x^2)^{3/2}} = \frac{3kQ}{8R^2}$$

$$\frac{8xR^2}{3} = (R^2 + x^2)^{3/2}$$

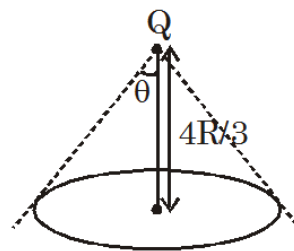
$$x = \frac{R}{\sqrt{2}} \rightarrow \frac{2\sqrt{2}R^3}{3} = \left(\frac{3R^2}{2}\right)^{3/2}$$

$$x = \frac{R}{3} \rightarrow \frac{8R^3}{9} = \left(R^2 + \frac{R^2}{9}\right)^{3/2}$$

$$x = \frac{R}{\sqrt{3}} \rightarrow \frac{8R^3}{3\sqrt{3}} = \left(R^2 + \frac{R^2}{9}\right)^{3/2} = \frac{8}{3\sqrt{3}}R^3$$

$$x = \frac{R}{2} \rightarrow \frac{4R^3}{3} = \left(R^2 + \frac{R^2}{4}\right)^{3/2}$$

3. Ans (0.05)



$$\theta = 37^\circ$$

$$\phi_{\text{flat}} = \frac{Q}{2\epsilon_0} (1 - \cos \theta)$$

$$= \frac{Q}{2\epsilon_0} \times \frac{1}{5} = \frac{Q}{10\epsilon_0}$$

$$\phi_{\text{quarter}} = 0 \text{ \& } \phi_{\text{vertical face}} = 0$$

$$\phi_{\text{base}} = \frac{\phi_{\text{flat}}}{2} = \frac{Q}{20\epsilon_0}$$

$$\phi_{\text{base}} + \phi_{\text{curved}} + \phi_{\text{vertical face}} = 0$$

$$\Rightarrow \phi_{\text{curved}} = -\frac{Q}{20\epsilon_0}$$

4. Ans (22.36)

$$\tan 45 = \frac{u \sin \theta - 10 \times 1}{u \cos \theta}$$

$$\frac{u \sin \theta}{g} = 2$$

$$u \sin \theta = 20$$

$$\Rightarrow u \cos \theta = 10$$

$$u = 10\sqrt{5} = 22.36$$

5. Ans (1.12 to 1.13)

$$\mu = \frac{\sin\left(\frac{60+46}{2}\right)}{\sin\left(\frac{60}{2}\right)} = \frac{4}{5} \times 2 = \frac{8}{5}$$

$$\frac{\mu}{\mu_0} = \frac{\sin\left(\frac{60+30}{2}\right)}{\sin\left(\frac{60}{2}\right)} = \sqrt{2}$$

$$\mu_0 = \frac{4\sqrt{2}}{5} = \frac{5.64}{5} = 1.13$$

6. Ans (11.00)

$$m_1 = -m_2$$

$$m = \frac{v}{u} = \frac{f}{u+f}$$

$$\frac{f}{-16+f} = -\frac{f}{-6+f}$$

$$-6+f = 16-f$$

$$2f = 22$$

$$f = 11 \text{ cm}$$

7. Ans (24.00)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_M}$$

$$\frac{1}{-60} - \frac{1}{40} = \frac{1}{f_M}$$

$$\frac{2+3}{-120} = \frac{1}{f_M} = -\frac{1}{24}$$

$$\frac{1}{f_M} = -\frac{2}{f_L} + \frac{1}{\infty}$$

$$\frac{1}{24} = 2(\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\Rightarrow R = 24 \text{ cm.}$$

8. Ans (4.00)

$$f_v - f_R = \omega f = 0.4 \times 10 = 4$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (5.00)

A, C, D, E, F

2. Ans (5.00)

A, E, F, G, H

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)

$$P(\alpha, \beta)$$

$$\text{coc is } \beta y = 2(x + \alpha)$$

$$y = \frac{2}{\beta}x + \frac{2\alpha}{\beta} \text{ is normal}$$

$$\frac{2\alpha}{\beta} = -2 \left(\frac{2}{\beta} \right) - \left(\frac{2}{\beta} \right)^3$$

$$\Rightarrow \alpha\beta^2 = -2\beta^2 - 4$$

$$\Rightarrow (\alpha + 2)\beta^2 + 4 = 0$$

$$\text{Locus : } (x + 2)y^2 + 4 = 0$$

$$(\alpha, \beta) \in I \quad \alpha = -2 - \frac{4}{\beta^2}$$

$$\therefore \beta^2 = 1, 4$$

$$\beta = 1, -1, 2, -2$$

$$\alpha = -6, -6, -3, -3$$

total 4 pairs

2. Ans (A,B,C)

$$T_n = 84 - 4n \text{ and}$$

$$S_{10} = \frac{10}{2} [2(80) + 9(-4)] = 620$$

$$S_{31} = \frac{31}{2} [2(80) + 30(-4)] = 620$$

$$T_{n-3} = 84 - 4(n-3)$$

$$96 - 4n$$

$$T_n \cdot T_{n-3} = (84 - 4n)(96 - 4n)$$

minimum occurs at 22.5

but n is natural

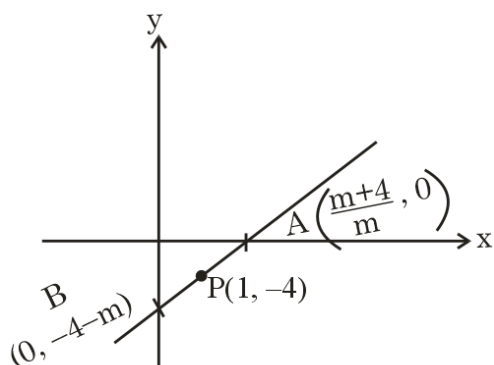
$$\therefore n = 22, 23$$

3. Ans (B,C,D)

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \text{ touches } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ at}$$

point $(x_1, -y_1)$. Also the locus of foot of perpendicular from the focus $(\sqrt{a^2 + b^2}, 0)$ to any tangent is the auxiliary circle $x^2 + y^2 = a^2$.

4. Ans (A,B,C)



Equation of AB is $y + 4 = m(x - 1)$

$$A = \text{Area} = \frac{1}{2} \left| \left(\frac{m+4}{m} \right) (-4-m) \right|$$

$$\Rightarrow (m+4)^2 = \pm 2Am$$

$$\Rightarrow (m+4)^2 = -2Am \text{ and } (m+4)^2 - 2mA = 0$$

$$\Rightarrow m^2 + m(8+2A) + 16 = 0$$

$$\& m^2 + m(8-2A) + 16 = 0$$

$$D = 4A^2 + 32A > 0, D = 4A^2 - 32A > 0$$

Now check the options.

5. Ans (A,B)

$$f(n) = \sum_{r=1}^n [r^2 ({}^nC_r - {}^nC_{r-1}) + (2r) {}^nC_r + {}^nC_r]$$

$$= \sum_{r=1}^n [(r^2 + 2r + 1) {}^nC_r - r^2 {}^nC_{r-1}]$$

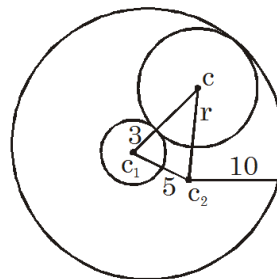
$$= \sum_{r=1}^n [(r+1)^2 {}^nC_r - r^2 {}^nC_{r-1}]$$

$$= \sum_{r=1}^n [V_{r+1} - V_r]$$

$$= V_2 - V_1 + V_3 - V_2 + \dots + V_{n+1} - V_n$$

$$= V_{n+1} - V_1 = (n+1)^2 {}^nC_n - 1 = n^2 + 2n.$$

6. Ans (A,C,D)



$$cc_2 = 10 - r$$

$$cc_1 = 3 + r$$

$$cc_1 + cc_2 = 13 > c_1c_2$$

$$(PS_1 + PS_2 = 2A > S_1S_2)$$

locus is ellipse

$$\text{so } 2a = 13 \text{ and } 2ae = 5$$

$$\Rightarrow e = \frac{5}{13}$$

Now,

$$b^2 = a^2(1 - e^2)$$

$$b^2 = \frac{169}{4} \times \left(1 - \frac{25}{169}\right)$$

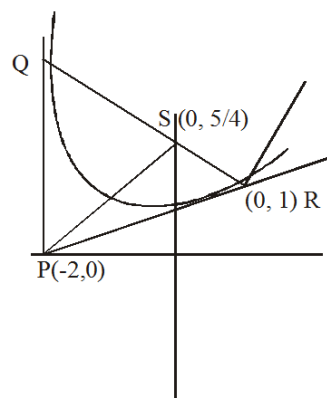
$$b = 6$$

$$A = \pi ab = 39\pi$$

SECTION-I (ii)

7. Ans (B,C,D)

$$f(x, y) = 0 \Rightarrow y = 1 + x^2$$



Since $\Delta PQS \sim \Delta RPS$

$$\therefore SQ \cdot SR = SP^2$$

$$= 4 + \frac{25}{16} = \frac{89}{16}$$

8. Ans (A,B,D)

By theory

$$k > \frac{3}{2} \therefore \text{Range} = \left(\frac{3}{2}, \infty\right)$$

9. Ans (C,D)

CONSONANTS: (H-1, M-1, R-1, T-1)

VOWELS : (A-1, I-2, O-1)

Sol. 1

CASE-I

All C+2 distinct V

$$({}^4C_4) ({}^3C_2) (6!) = 2160$$

CASE-II

All C and 2I

$$({}^4C_4) ({}^2C_2) \left(\frac{6!}{2!} \right) = 360$$

CASE-III

3 C and 3 distinct V

$$({}^4C_3) ({}^3C_3) (6!) = 2880$$

CASE-IV

3 C and 3 V (2I+A/O)

$$({}^4C_3) ({}^2C_1) \left(\frac{6!}{2!} \right) = 2880$$

CASE-V

ALL V and 2 C

$$({}^4C_2) \left(\frac{6!}{2!} \right) = 2160$$

Sol:2

CASE-I

All C+2 distinct V

$$({}^4C_4) ({}^3C_2) \left(\frac{6!}{2!} \right) = 1080$$

CASE-II

All C and 2I

$$({}^4C_4) ({}^2C_2) \left(\frac{6!}{2!} \right) = 360$$

CASE-III

3 C and 3 distinct V

$$({}^4C_3) ({}^3C_3) \left(\frac{6!}{3!} \right) = 480$$

CASE-IV

3 C and 3 V (2I+A/O)

$$({}^4C_3) ({}^2C_1) \left(\frac{6!}{2! \times 3} \right) = 960$$

CASE-V

ALL V and 2 C

$$({}^4C_2) \left(\frac{6!}{2! \times 12} \right) = 180$$

10. Ans (A,C)

CONSONANTS: (H-1, M-1, R-1, T-1)

VOWELS : (A-1, I-2, O-1)

Sol. 1

CASE-I

All C+2 distinct V

$$({}^4C_4) ({}^3C_2) (6!) = 2160$$

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All C and 2I

$$({}^4C_4) ({}^2C_2) \left(\frac{6!}{2!} \right) = 360$$

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3 C and 3 distinct V

$$({}^4C_3) ({}^3C_3) (6!) = 2880$$

CASE-IV

3 C and 3 V (2I+A/O)

$$({}^4C_3) ({}^2C_1) \left(\frac{6!}{2!} \right) = 2880$$

CASE-V

ALL V and 2 C

$$({}^4C_2) \left(\frac{6!}{2!} \right) = 2160$$

Sol:2

CASE-I

All C+2 distinct V

$$({}^4C_4) ({}^3C_2) \left(\frac{6!}{2!} \right) = 1080$$

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$$({}^4C_3) ({}^3C_3) \left(\frac{6!}{3!} \right) = 480$$

CASE-IV

3 C and 3 V (2I+A/O)

$$({}^4C_3) ({}^2C_1) \left(\frac{6!}{2! \times 3} \right) = 960$$

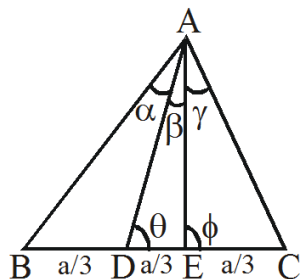
CASE-V

ALL V and 2 C

$$({}^4C_2) \left(\frac{6!}{2! \times 12} \right) = 180$$

SECTION-II

1. Ans (1.33)



By m - n theorem in $\triangle ABE$

$$(1 + 1) \cot \theta = 1. \cot \alpha - 1. \cot \beta \quad \dots(1)$$

and in $\triangle ABC$

$$(1 + 1) \cot \phi = 1 \cot \beta - 1. \cot \gamma \quad \dots(2)$$

$$x = \cot \alpha, y = \cot \beta, z = \cot \gamma$$

$$y - z = \cot \phi = 2 \cot(\theta + \beta)$$

$$= 2 \left(\frac{\cot \theta \cot \beta - 1}{\cot \theta + \cot \beta} \right)$$

$$y - z = 2 \left(\frac{\left(\frac{x-y}{2} \right) y - 1}{\left(\frac{x-y}{2} \right) + y} \right)$$

$$(y - z) = \frac{2(xy - y^2 - 2)}{(x - y)}$$

$$xy + y^2 - xz = 2xy - 2y^2 - 4$$

$$(x + y)(y + z) = 4(1 + y^2)$$

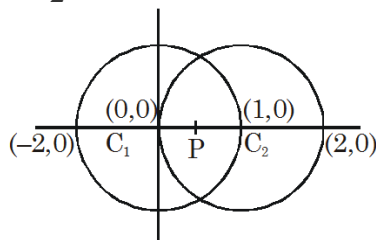
$$\frac{(x + y)(y + z)}{1 + y^2} = 4$$

2. Ans (2.25)

$$4 \left\{ \left(x_1 - \frac{1}{2} \right)^2 + y_1^2 + \left(x_2 - \frac{1}{2} \right)^2 + y_2^2 \right\} \in [a, b]$$

minimum distance of P $\left(\frac{1}{2}, 0 \right)$ from given circles

are $\frac{1}{2}$ and maximum distance are $\frac{3}{2}$



$$a = 4 \left[\frac{1}{4} + \frac{1}{4} \right] = 2$$

$$b = 4 \left[\frac{9}{4} + \frac{9}{4} \right] = 18$$

3. Ans (0.80)

$$a_n = \frac{(2n + 1)^9 C_{n+1}}{9} \quad \text{and}$$

$$b_n = \frac{(3n + 1)^8 C_{n+1}}{8}$$

a_n is max when $n + 1 = 5$ and b_n is min when $n = 0$

$$\Rightarrow p = 4 \text{ and } q = 0$$

$$p + q = 4$$

4. Ans (2.33)

$$\frac{\sec 8\theta - 1}{\sec 4\theta - 1} = \frac{(1 - \cos 8\theta)(\cos 4\theta)}{(1 - \cos 4\theta)(\cos 8\theta)}$$

$$= \frac{2\sin^2 4\theta \cdot \cos 4\theta}{2\sin^2 2\theta \cdot \cos 8\theta} = \frac{\tan 8\theta \cdot 2 \sin 2\theta \cos 2\theta}{2\sin^2 2\theta}$$

$$= \frac{\tan 8\theta}{\tan 2\theta} = \frac{2 \tan 4\theta}{(1 - \tan^2 4\theta) \cdot \tan 2\theta}$$

$$\left(\text{Where } \tan 4\theta = \frac{2 \tan 2\theta}{1 - \tan^2 2\theta} \right)$$

$$\left(\frac{4 \tan 2\theta - 4 \tan^3 2\theta}{1 - 6 \tan^2 2\theta + \tan^4 2\theta} \right) \frac{1}{\tan 2\theta}$$

$$\frac{4 - 4 \tan^2 2\theta}{1 - 6 \tan^2 2\theta + \tan^4 2\theta}$$

$$a = 4, b = -4, c = -6, d = 1$$

5. Ans (1.75)

a, b, c are in H.P.

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in A.P.}$$

$$4acx + y(ab + bc + ca - abc) + abc = 0$$

$$\Rightarrow \frac{4}{b}x + y \left(\frac{1}{c} + \frac{1}{a} + \frac{1}{b} - 1 \right) + 1 = 0$$

$$\frac{4}{b}x + y \left(\frac{3}{b} - 1 \right) + 1 = 0$$

$$\frac{1}{b}(4x + 3y) + (1 - y) = 0$$

$$(1 - y) + \lambda(4x + 3y) = 0 \quad (\lambda \in \mathbb{R}^+)$$

$$L_1 \equiv 1 - y = 0 \Rightarrow y = 1$$

$$L_2 \equiv 4x + 3y = 0 \Rightarrow x = -\frac{3}{4}$$

$$\text{point } \left(-\frac{3}{4}, 1 \right) \text{ lies on } x - y = k \text{ so } k = -\frac{7}{4}$$

6. Ans (6.00)

For consistency,

$$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$$

$$\Rightarrow (a-b)(a-b^2) = 0$$

$\therefore$ Possible ordered pairs are

(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (4, 2)

7. Ans (1.80)

$$\frac{a}{b} = \frac{b}{c} = \lambda$$

$$\Rightarrow a = \lambda b$$

$$b = \lambda c \Rightarrow a = \lambda^2 c$$

$$E = \frac{\log(abc^4)}{\log ab} \cdot \frac{\log(a^4 bc)}{\log bc}$$

$$= \frac{\log(\lambda c^2)^3 \cdot \log(\lambda^3 c^2)^3}{\log(\lambda^3 c^2) \cdot \log(\lambda c^2)}$$

$$\Rightarrow E = 3 \times 3 = 9$$

8. Ans (0.50)

$$\prod_{s=0}^{n-1} (2 \cos(2^s x) - 1)$$

$$\Rightarrow (2 \cos x - 1)(2 \cos 2x - 1)(2 \cos 2^2 x - 1) \dots (2 \cos 2^{n-1} x - 1)$$

$$\Rightarrow f_n(x) = \frac{2 \cos 2^n x + 1}{2 \cos x + 1}$$

$$f_5(x) = \frac{2 \cos 32x + 1}{2 \cos x + 1}$$

$$f_5\left(\frac{2\pi}{31}\right) = \frac{2 \cos \frac{64\pi}{31} + 1}{2 \cos \frac{2\pi}{31} + 1} = 1$$